

Grade 6 Accelerated
Unit 14
Lesson 9 Homework

Name Answer Key
Date _____
Period _____

1. For each inequality, write three numbers that could represent the solution set.

Answer may vary.

A. $x > 9$ 10, 11, 12

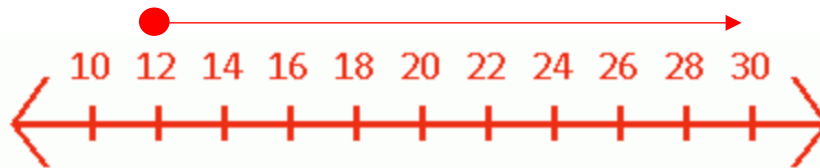
B. $-12 \leq m$ -11, -9, -7

C. $g < 0$ -1, -5, -10

2. The Overseas Highway, also known as the "Highway that Goes to Sea," runs 113 miles from Miami down to Key West. The road connects the Florida Keys to the Florida mainland. The speed limit of the road is 55 miles per hour. Represent the speeds that one could drive without exceeding the speed limit in an inequality with s representing the speed.

$s \leq 55$ mph

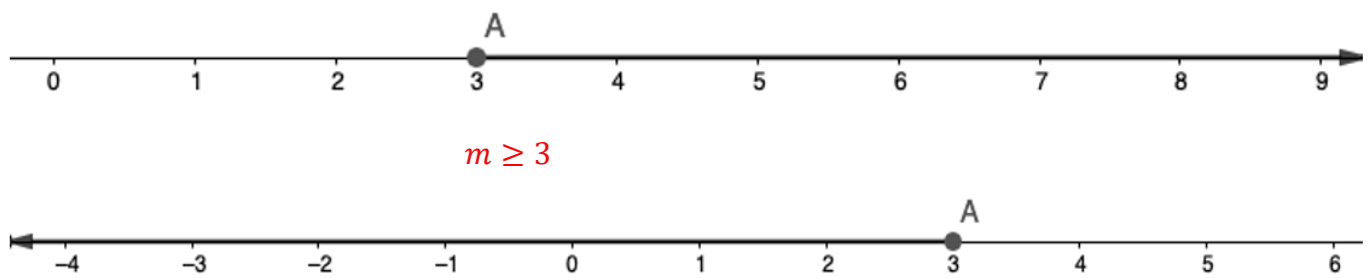
3. Graph the solution $x \geq 12$.



4. Chris said that the numbers $(-9, -10, -11)$ be in the solution set for the inequality $r < -9$. Explain if Chris' answer is correct.

-9 is incorrect because the inequality says r is "less than -9 , not "less than or equal to."

5. Compare and contrast the two number lines.



3 is part of the solution set of both inequalities.

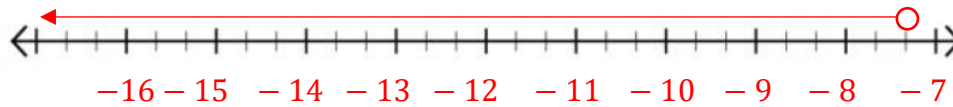
$m \leq 3$

6. Use the inequality $-9.75 > y - 2.5$ to complete the following:

A. Find the solution set by isolating the variable. Show your work.

$$\begin{array}{r} -9.75 > y - 2.5 \\ +2.5 \quad +2.5 \\ \hline -7.25 > y \quad \text{or} \quad y < -7.25 \end{array}$$

B. Graph the solution set on the number line.



C. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

$$\begin{array}{ll} -9.75 > -10 - 2.5 & -9.75 > -16 - 2.5 \\ -9.75 > -12.5 \text{ T} & -9.75 > -18.5 \text{ T} \end{array}$$

7. How do you isolate a variable when solving an inequality?
Use the opposite operations just as you would when solving an equation.

Solve each inequality. Then, graph its solution set.

8. $-15\frac{1}{4} \leq m + 6.5$

$$\begin{array}{r} -6.5 \quad -6.5 \\ -21.75 \leq m \\ m \geq -21.75 \end{array}$$



9. $48 > y - 24$

$$\begin{array}{r} +24 \quad +24 \\ 72 > y \\ y < 72 \end{array}$$



10. $21.4 + w < 56.9$

$$\begin{array}{r} -21.4 \quad -21.4 \\ w < 35.5 \end{array}$$

